CITY MEDICAL CENTER

Laboratory Investigation Report

Patient Name:	John Smith	**Patient ID:**	P001
Date of Birth:	March 15, 1979	**Gender:**	Male
Ordering Doctor:	Dr. Mehta	**Specialty:**	Nephrologist
Sample Collected:	January 6, 2025	**Report Date:**	January 7, 2025
Clinical Notes:	Type 2 Diabetes followup.		

1. Complete Blood Count (CBC)

Test	Result	Reference Range	Unit	Status
Hemoglobin	13.8	12.0 - 16.0	g/dL	Normal
WBC Count	9.2	4.0 - 11.0	x10³/µL	Normal
RBC Count	4.5	3.8 - 5.2	x10■/µL	Normal
Platelets	220	150 - 400	x10³/µL	Normal
Neutrophils	68	40 - 75	%	Normal
Lymphocytes	24	20 - 45	%	Normal
Eosinophils	6.2	1 - 4	%	HIGH ■
Hematocrit	41.2	36 - 46	%	Normal
MCV	88.4	80 - 100	fL	Normal

2. Thyroid Function Tests

Test	Result	Reference Range	Unit	Status
TSH	2.1	0.4 - 4.0	mIU/L	Normal
Free T4	1.2	0.8 - 1.8	ng/dL	Normal
Free T3	3.4	2.3 - 4.2	pg/mL	Normal

3. Allergy & Pulmonary Markers

Test	Result	Reference Range	Unit	Status
Total IgE	420	< 100	IU/mL	HIGH ■
Specific IgE (Dust mites)	3.8	< 0.35	kUA/L	HIGH ■
Specific IgE (Pollen)	2.1	< 0.35	kUA/L	HIGH ■
CRP (C-Reactive Protein)	8.4	< 5.0	mg/L	HIGH ■
Peak Expiratory Flow	310	380 - 500	L/min	LOW ■

4. Basic Metabolic Panel

Test	Result	Reference Range	Unit	Status
Glucose (Fasting)	94	70 - 100	mg/dL	Normal
Sodium	139	136 - 145	mmol/L	Normal
Potassium	3.9	3.5 - 5.1	mmol/L	Normal
Creatinine	0.78	0.5 - 1.1	mg/dL	Normal
eGFR	88	> 60	mL/min/1.73m²	Normal
ALT	22	7 - 40	U/L	Normal
AST	19	10 - 40	U/L	Normal

5. Interpreting Physician Notes

Dr. Sharma (Pulmonologist) — January 7, 2025

Key Findings:
- Elevated eosinophils (6.2%) suggest allergic inflammation, consistent with asthma.
- Markedly elevated Total IgE (420 IU/mL) and positive specific IgE to dust mites and pollen strongly indicate allergic sensitization as a trigger for asthma symptoms.
- CRP mildly elevated at 8.4 mg/L — suggests low-grade airway inflammation.
- Peak expiratory flow reduced at 310 L/min (below expected 380-500 L/min range) — indicates moderate airflow limitation at time of testing.
- Thyroid function normal — Levothyroxine dose appears adequate.
- Kidney and liver function within normal limits.

Recommendations:
1. Consider allergen immunotherapy referral given high IgE and specific sensitization.
2. Review inhaler technique and step up to combination ICS/LABA if not already done.
3. Avoid known allergens (dust mites, pollen) — environmental control measures advised.
4. Repeat PEF measurement after bronchodilator to assess reversibility.
5. Follow up in 4 weeks to reassess respiratory status.

Signed: Dr. Priya Sharma, MD | Pulmonology | City Medical Center